SOTHEA HAS

PhD in Applied Mathematics

Summary

Lecturer-Researcher, and Head of Research and Data Analytics Lab (ReDA Lab) at ITC. Expert in Machine Learning, Statistics, and Data Science. Currently leading Engineering, Master's, and PhD students in applying AI solutions to the Education and Agriculture sectors. Research background includes Predictive Modeling via Aggregation and Deep Learning for atmospheric gravity waves (Schmidt Sciences, DataWave project).

Experience

Projects (member & Co-PI)

- **01/2026 - 12/2027: Co-PI of HEIP 2 Project at ITC (Grant Awarded), Cambodia.**
- **01/09/2025 - 30/08/2026: Co-PI of LBE Project at ITC, Cambodia.**
- **03/2024 - 08/2024**: **Member of Data Wave Project of Schmidt Sciences** for gravity wave modeling (Poster presented at **University of Cambridge,** July 2024).
- **09/2022 - 02/2024**: **Member of ANR Project BOOST3R (ANR-17-CE01-0016-01).**

Postdoctoral researcher — Sep 2022 - August 2024

Laboratoire de Météorologie Dynamique - ENS

- Reconstructing balloon-observed gravity wave momentum fluxes using Machine Learning and inputs from ERA5.
- Analyzing **SGD** algorithm using continuous-time **stochastic processes**.

Teaching

Institute of Technology of Cambodia (ITC) — Sep 2024 - Present

- Statistics (Year 3)
- Exploratory Data Analysis & Unsupervised Learning (Year 5)
- Advanced Machine Learning (Master 2)
- EDA & Unsupervised Learning (Master 1)

Lecturer Researcher

UFR Mathématiques - Paris 7 — Sep 2018 - Mar 2024

- Data Analysis with **R** and **Rstudio**.
- Data Mining with **R** and **Rstudio**.
- Exploratory Data Analysis with **R** and **Rstudio**.
- Algorithm and Programming with **Python**.
- Big Data Technologies with **Python** and **Spark**.
- Statistical Inference and Data Modeling.

Teaching Assistant & ATER

PhD research

LPSM - Sobronne Université — Aug 2018 - Aug 2022

- Predictive Models based on Aggregation Methods in Machine Learning.
- Developed "**gradientcobra**" python library for aggregation method in regression.

Publication

- Estimating balloon-observed gravity wave momentum fluxes using ML & input from ERA5. *Published in JGR - Atmosphere*, **Has et al.** (**2024**).
- Gradient COBRA: A kernel-based consensual aggregation for regression. *Published in Journal of Data Science, Statistics and Visualization*, **S. Has (2023)**.
- A consensual aggregation of randomly projected high-dimensional features of predictions. *Available in HAL*, **S. HAS** (**2022**).
- Machine learning methods applied to the global modeling of event-driven pitch angle diffusion coefficients during high-speed streams. *Published in Frontiers Physics*, **Kluth et al. (2022), co-author**.
- KFC: A clusterwise supervised learning procedure based on aggregation of distances. *Published in Journal of Statistical Computation and Simulation*, **Has et al. (2021)**.

Contact

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Phnom Penh, Cambodia

Education

Postdoctoral researcher

CNRS and ENS, Paris, France
Topic: Applied ML in Atmospheric Science
2024

PhD in Applied Math

Sorbonne Université, Paris, France
Topic: Theoretical & Applied ML
2022

Master M2MO

Université Paris Diderot, Paris, France
Main focus: Financial Math & Data Science
2018

Engineer of Applied Math

ENSIIE, Evry, France
Main focus: Financial Math & Data Science
2018

Bachelor of Pure Math

Royal University of Phnom Penh, Cambodia
Main focus: Pure mathematics
2014

Language

Khmer: mother tongue

English: fluent

French: conversational

Technical skills

Languages: Python, R, SQL, MatLab, C++

Libraries: PyTorch, TensorFlow, Sklearn, Pandas, Numpy, Scipy, PySpark, GradientCOBRA...